

Udemy Data Analytics Project: Exploring Course Performance and Revenue Opportunities

Data-driven decision-making plays a major role in how online learning platforms grow and stay competitive. In this project, I explored Udemy course data across multiple subject areas to understand course performance and identify potential opportunities to improve revenue.

This project was done as a **learning-based analysis**, using publicly available Udemy course data.

Project Overview

The objective of this analysis was to evaluate how different Udemy courses perform based on pricing, subscriptions, ratings, content duration, and revenue. By comparing subjects such as Web Development, Business Finance, Graphic Design, and Musical Instruments, I aimed to uncover trends that could help guide business decisions.

Business Context

In this scenario, I worked from the perspective of a data analyst supporting an education technology platform. The management team needed insights on which course categories were driving the most revenue and which areas required improvement.

One assumption explored during the analysis was whether Web Development courses, which are generally popular, could potentially be priced higher to increase overall earnings. The final goal was to provide clear, data-backed recommendations that could support revenue growth in upcoming quarters.

Tools Used

- **Microsoft Excel** – Data cleaning and preprocessing using Power Query
- **Microsoft SQL Server** – Writing queries to generate insights
- **Microsoft Power BI** – Creating dashboards and visualizations

Data Cleaning Process

Before analysis, the dataset was cleaned and prepared in Excel. The main steps included:

- Splitting the published timestamp into separate date and time fields
- Correcting data types for all columns

- Creating a new column to classify courses as *Free* or *Paid*
- Removing unnecessary columns such as URLs
- Renaming headers for clarity and consistency
- Removing duplicate and blank records

These steps ensured the dataset was accurate, consistent, and ready for analysis.

Below are the data cleaning processes I carried out using the power query editor in Microsoft Office Excel:

1. split the “published timestamp” column using the split column by delimiter tool.

Published timestamp column with date and time

Transform		Combine		Parameters		Data Sources		New	
t", [PromoteAllScalars=true])									
ing	ABC 123	content_duration	ABC 123	published_timestamp	ABC 123	subject			
	0.56		8	2013-04-20T02:25:22Z		Business Finance			
	0.95		1.5	2013-04-07T21:39:25Z		Business Finance			
	0.91		1.5	2013-12-25T19:53:34Z		Business Finance			
	0.18		1.5	2014-05-27T16:22:16Z		Business Finance			
	0.37		10	2016-01-21T01:38:48Z		Business Finance			
	0.89		3.5	2013-10-16T17:40:23Z		Business Finance			

Split column using the delimiter “T”

Refresh
Preview
Manage

Choose
Columns
Remove
Columns
Keep
Rows
Remove
Rows

Split
Column
Group
By
Replace
Values

Split Column by Delimiter

Specify the delimiter used to split the text column.

Select or enter delimiter

--Custom--

T

Split at

☐ Left-most delimiter

☐ Right-most delimiter

☒ Each occurrence of the delimiter

Advanced options

Quote Character

"

☐ Split using special characters

Insert special character

OK

Cancel

2. I changed the data type for each column to their correct data formats.

Made the Course Title column a text (string) data type.

	course_id	course_title	
1	49798	1.2	Decimal Number
2	48841	\$	Currency
3	133536	123	Whole Number
4	151668	%	Percentage
5	648826		Date/Time
6	97338		Date
7	321410		Time
8	301442		Date/Time/Timezone
9	189160		Duration
10	191854		Text
11	754028		True/False
12	265960		Binary
13	308690		
14	285638		

3. I added a conditional column which tells whether a course is “Free or Paid”. I used the data in the price column to populate this column.

	course_id	course_title	price	Free/Paid
1	49798	1.2	100.00	Paid
2	48841	\$	200.00	Paid
3	133536	123	21442	Paid
4	151668	%	2118	Paid
5	648826			
6	97338			
7	321410			
8	301442			
9	189160			
10	191854			
11	754028			
12	265960			
13	308690			
14	285638			

If price = 0 output will be “Free”, else “Paid”

4. I deleted irrelevant columns that would not be useful for my analysis.

5. I ensured that I had clear and concise names for headers and used dashes or underscores in between words to make it easier to parse later on.

6. I removed duplicate rows of data.

7. I removed blank rows.

SQL Analysis

After cleaning the data, it was imported into SQL Server Management Studio. Using SQL:

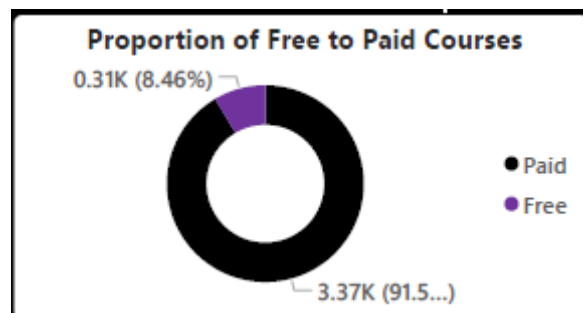
- Multiple datasets were combined using the **UNION** operator
- A new **Revenue** column was created using the formula:
Revenue = Price × Number of Subscribers
- Queries were written to analyze revenue, subscriber counts, ratings, course levels, and subject-wise performance

Power BI & DAX

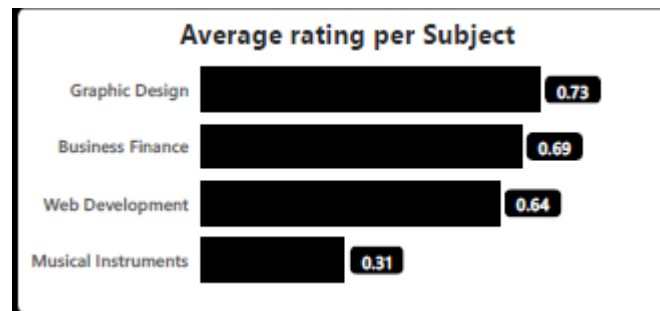
Power BI was used to visualize insights clearly and interactively. Key steps included:

- Creating a calendar table using DAX (Year, Month, Quarter)
- Building measures for revenue, averages, and counts
- Designing dashboards to highlight trends and comparisons across subjects

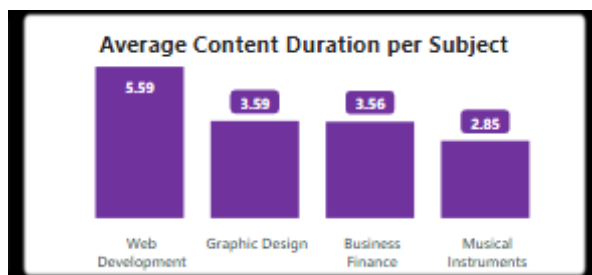
From a total of 3,676 courses, 3,365 (91.54%) are paid courses while 311 (8.46%) are free courses.



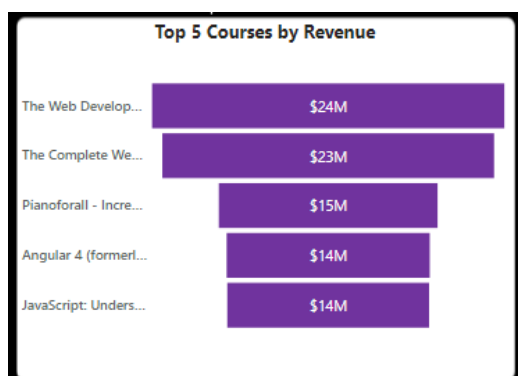
- As regards subject ratings, graphic design has the highest average rating with a value of 0.73, followed by business finance with an average rating of 0.69. Musical instruments have the lowest average rating of 0.31.



- Web Development on an average has the highest content duration with an average value of 5.59 minutes, while Musical Instrument has the lowest average content duration with a value of 2.85 minutes.

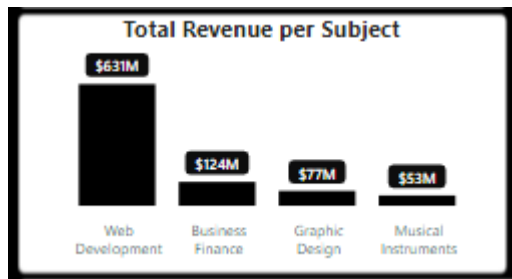


- Four out of the top 5 courses that generated the highest revenue are courses under the subject Web Development, the third highest revenue generating course is a course under the subject Musical Instruments.

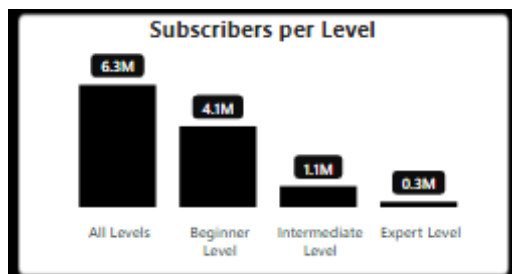


- Web Development generated the highest total revenue with a value of \$631,082,465, while Musical Instruments generated the lowest total revenue with a value of

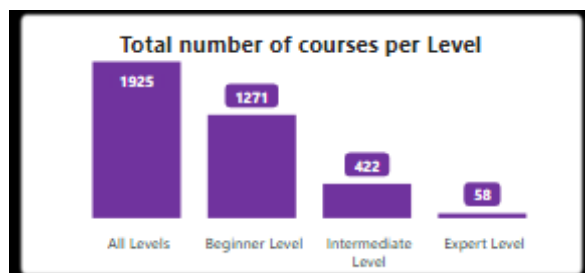
\$53,359,055.



- Courses in the All Levels category have the highest number of subscribers, followed by Beginner level courses. Expert level courses have the smallest amount of subscribers.

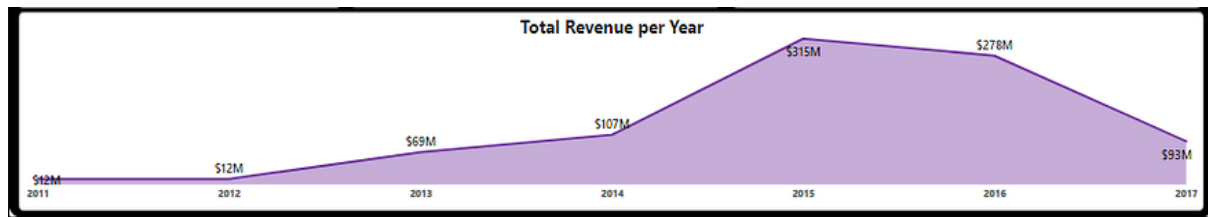


- Of the 3,676 available in our database, there are a total of 1,925 All Levels courses, 1,271 Beginner Level courses, 422 Intermediate Level courses, and 58 Expert level courses.



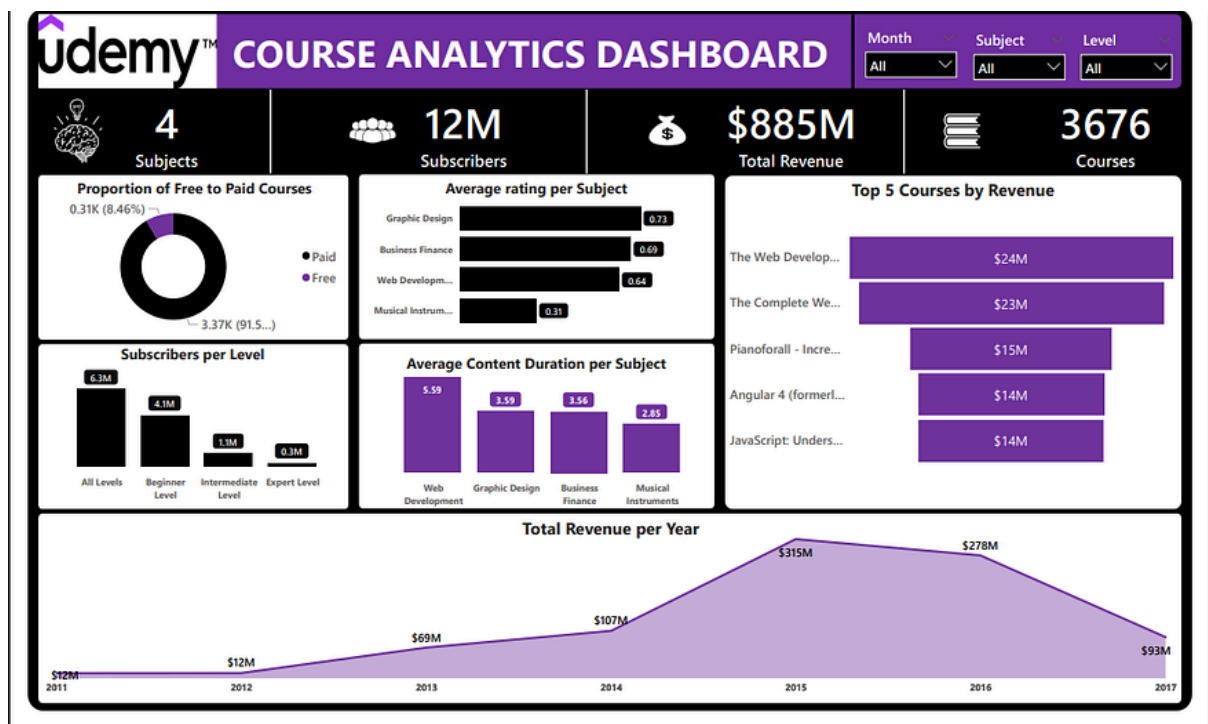
- Revenue over the years grew steadily from 2011 (\$11,643,429) to 2015 (\$314,510,395) and dropped slightly in 2016 to a value of \$277,694,340, but there was a huge drop in revenue generation between the years 2016 and 2017. In 2017 a total revenue of \$92,970,175 was

generated.



Power BI Visualization:

Press enter or click to view image in full size



Key Insights

- Out of 3,676 courses, around 91.5% are paid, while 8.5% are free
- **Graphic Design** has the highest average course rating
- **Musical Instruments** has the lowest average rating and lowest revenue
- **Web Development** courses have the longest average content duration
- **Web Development** generates the highest total revenue overall

- Most subscribers are enrolled in **All Levels** and **Beginner** courses
- **Expert-level courses** are very limited compared to other levels
- Revenue increased steadily from 2011 to 2015, declined slightly in 2016, and dropped significantly in 2017

Recommendations

Based on the analysis, the following actions could help improve revenue:

1. Convert more high-performing free courses into paid offerings
2. Improve the quality and structure of Musical Instrument courses to boost engagement
3. Review Graphic Design courses, as ratings are strong but revenue is relatively low
4. Increase the number of Intermediate and Expert-level courses
5. Continue investing in Web Development courses due to their strong revenue performance

Conclusion

This project helped me practice the full data analysis workflow — from data cleaning and SQL querying to visualization and business recommendations. It reinforced the importance of using data to support strategic decisions and provided hands-on experience with Excel, SQL, and Power BI.

Note: This project is a learning-based recreation inspired by publicly available Udemy course datasets and data analytics case studies. The analysis and presentation were rewritten and practiced independently for skill development.